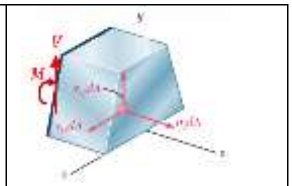
SHEAR STRESS IN BEAM DUE TO TRANSVERSE LOADS

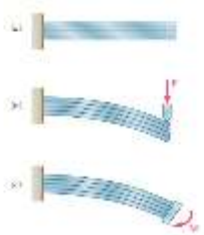
A transverse load applied to a beam will result in normal stress and shear stress in any given cross section. The normal stresses are caused by the bending moment M in that section and the shear stress by the shear force V



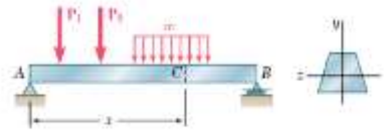
Due to the complimentary property of shear, an equal shear stress will be acting in the horizontal planes also. This can be seen in the following pictures below. Since the top and bottom surfaces are free of any shear force, the shear force on the cross section near to the top and bottom surface should also be zero (satisfying the boundary conditions).

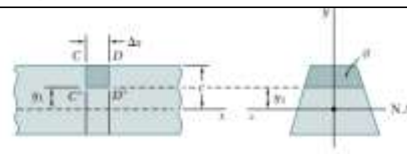
Boards not bonded together

Boards bonded together



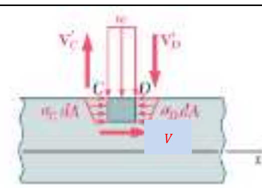
Consider a prismatic beam AB with a vertical plane of symmetry that supports various transverse loads as shown. At a distance x from A, an element $CC'D'D$ of length dx extending across the width is cut out from the beam. The bottom surface of the element is at a distance y from the neutral axis.

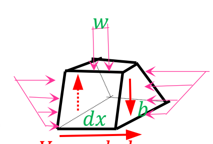




Considering the force equilibrium in the horizontal direction, we can write

$$\int_{\sigma} \sigma_C \, dA + V - \int_{\sigma} \sigma_D \, dA = 0$$





$$\int_{y=y_1}^{y=c} \frac{M_z^C}{I_{zz}} y dA + V - \int_{y=y_1}^{y=c} \frac{M_z^D}{I_{zz}} y dA = 0$$

$$\tau \, b.dx = \int_{y=y_1}^{y=c} \frac{M_z^D - M_z^C}{I_{zz}} y dA \quad \gg\gg \quad \tau = \frac{1}{I_{zz} b}\left(\frac{dM}{dx}\right)\int_{y=y_1}^{y=c} y \, dA \gg \tau = \frac{V}{I_{zz} b}[A'\bar{y}']_{y=y_1}^{y=c}$$

$$\tau = \frac{VQ}{bI_{zz}}$$

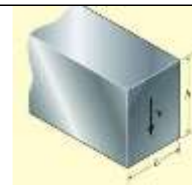
$[A'\bar{y}']_{y=y_1}^{y=c}$ is the first moment of the cross-sectional area from y_1 to c $(outer\ surface)$ with respect to the neutral axis.

Important Points

- Shear forces in beams cause *nonlinear shear-strain* distributions over the cross section, causing it to *warp*.
- Due to the complementary property of shear stress, the shear stress developed in a beam acts over the cross section of the beam and along its longitudinal planes.
- The *shear formula* was derived by considering horizontal force equilibrium of the longitudinal shear-stress and bending-stress distributions acting on a portion of a differential segment of the beam.
- The shear formula is to be used on straight prismatic members made of homogeneous material that has linear elastic behavior. Also, the internal resultant shear force must be directed along an axis of symmetry for the cross-sectional area.
- The shear formula should not be used to determine the shear stress on cross sections that are short or flat, at points of sudden cross-sectional changes, or at a point on an inclined boundary.

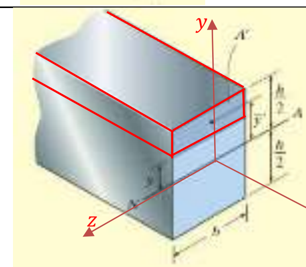
#Example 1

Determine the shear stress distribution across the rectangular cross section of a beam



Consider a horizontal section of the beam at a distance y from the neutral axis. The depth of the element is $\left(\frac{h}{2}-y\right)$ and the cross-sectional area of the element $A'=b\left(\frac{h}{2}-y\right)$. The centroid of this area from the y-axis,

$$\bar{y}'=y+\frac{\frac{h}{2}-y}{2}=\frac{y}{2}+\frac{h}{4}=\frac{1}{2}\left(\frac{h}{2}+y\right)$$



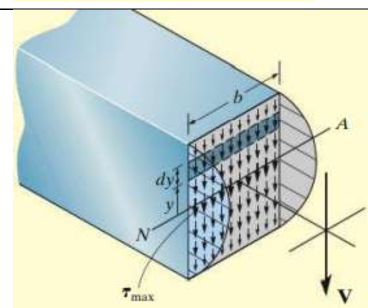
$$A'\bar{y}'=b\left(\frac{h}{2}-y\right)\frac{1}{2}\left(\frac{h}{2}+y\right)=\frac{b}{2}\left(\frac{h^2}{4}-y^2\right)$$

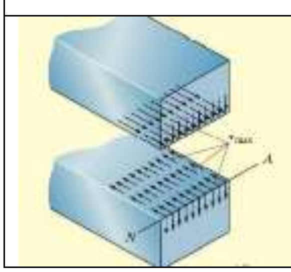
The shear stress on this horizontal plane

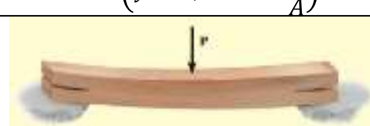
$$\tau=\frac{V}{I_{zz}b}A'\bar{y}'=\frac{V}{\left(\frac{bh^3}{12}\right)b}\times\frac{b}{2}\left(\frac{h^2}{4}-y^2\right)=\frac{6V}{bh^3}\left(\frac{h^2}{4}-y^2\right)$$

The shear stress distribution is parabolic.

$$\tau=\left(\frac{V}{bh}\right)\left(\frac{3}{2}-\frac{6y^2}{h^2}\right)\begin{Bmatrix}y=\pm\frac{h}{2},\ \ \tau=0\\ y=0,\tau=1.5\ \frac{V}{A}\end{Bmatrix}$$







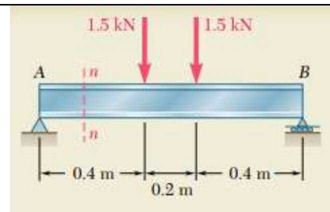
$$\int_A \tau\, dA=\int_{-h/2}^{h/2}\frac{6V}{bh^3}\left(\frac{h^2}{4}-y^2\right)b\,dy$$

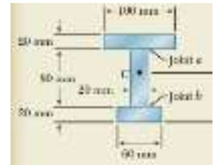
$$=\frac{6V}{h^3}\left[\frac{h^2}{4}y-\frac{1}{3}y^3\right]_{-h/2}^{h/2}$$

$$=\frac{6V}{h^3}\left[\frac{h^2}{4}\left(\frac{h}{2}+\frac{h}{2}\right)-\frac{1}{3}\left(\frac{h^3}{8}+\frac{h^3}{8}\right)\right]=V$$

#Example 2

Beam AB is made of three planks glued together and is subjected, in its plane of symmetry, to the loading shown. Knowing that the width of each glued joint is 20 mm, determine the average shearing stress in each joint n-n of the beam



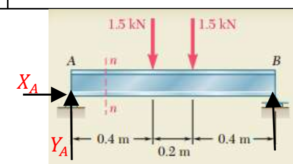


$\Sigma F_x = 0 \ggg X_A = 0$

$\Sigma M_A = 0 \ggg$

$R_B \times 1m - 1.5\ kN \times 0.4m - 1.5\ kN \times 0.6m = 0 \gg R_B = 1.5\ kN$

$\Sigma F_y = 0 \ggg R_B + Y_A - 1.5\ kN - 1.5\ kN = 0 \ggg Y_A = 1.5\ kN$

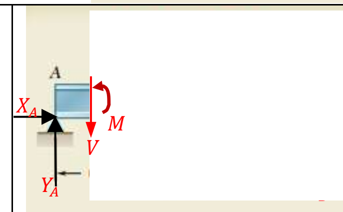


Section n-n

$$Y_A - V = 0 \ggg V = 1.5\ kN$$

	A_i mm^2	y_i mm	$A_i y_i$
1.	$20 \times 100 = 2000$	110	220000
2.	$20 \times 80 = 1600$	60	96000
2.	$20 \times 60 = 1200$	10	12000
	$\Sigma A_i = 4800$		$\Sigma A_i y_i = 328000$

$$\bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = 68.3\ mm$$

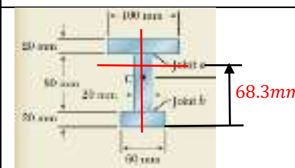


Moment of inertia of the cross-section about the neutral axis (centroidal z-axis)

$$I_{zz} = \left\{\frac{100 \times 20^3}{12} + 2000 \times (110 - 68.3)^2\right\} + \left\{\frac{20 \times 80^3}{12} + 1600 \times (68.3 - 60)^2\right\} + \left\{\frac{60 \times 20^3}{12} + 1200 \times (68.3 - 10)^2\right\}$$

$$I_{zz} = \{66666.7 + 3477780\} + \{853333.3 + 110224\} + \{40000 + 4078668\}$$

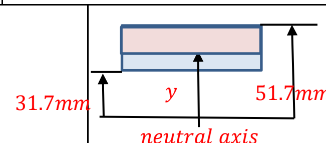
$$I_{zz} = 8626.7 \times 10^3\ mm^4 = 8.63 \times 10^{-6}\ m^4$$



The shear stress on the cross section of top flange

$A' = 0.1(0.0517 - y) \lll\ggg \bar{y}' = \frac{(0.0517 + y)}{2}$

$$A'\bar{y}' = 0.05(0.0517^2 - y^2)$$



At the top of the flange, $y = 0.0517\ m$. Therefore, $\tau = \frac{V}{I_{zz}b}A'\bar{y}' = 0$

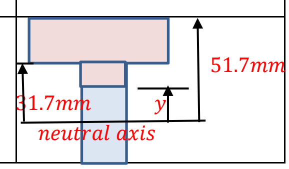
At the bottom of the top flange, $y = 0.0317\ m \ggg\gg A'\bar{y}' = 8.34 \times 10^{-5}$

$$\tau = \frac{V}{I_{zz}b}A'\bar{y}' = \frac{1.5 \times 10^3}{8.63 \times 10^{-6}\ m^4 \times 0.1\ m} \times 8.34 \times 10^{-5} = 145\ kPa$$

Shear stress on the cross section of the web

$$A'\bar{y}' = (0.1 \times 0.02)(0.417) + 0.02(0.0317 - y)\left(\frac{0.0317 + y}{2}\right)$$

$$= 8.34 \times 10^{-5} + 0.01(0.0317^2 - y^2)$$



Top of the web (junction), $y = 0.0317\ m \ggg\gg A'\bar{y}' = 8.34 \times 10^{-5}$

$$\tau = \frac{V}{I_{zz}b}A'\bar{y}' = \frac{1.5 \times 10^3}{8.63 \times 10^{-6}\ m^4 \times 0.02\ m} \times 8.34 \times 10^{-5} = 725\ kPa$$

At the neutral surface, $y = 0 \ggg\gg A'\bar{y}' = 9.345 \times 10^{-5}$

$$\tau = \frac{V}{I_{zz}b}A'\bar{y}' = \frac{1.5 \times 10^3}{8.63 \times 10^{-6}\ m^4 \times 0.02\ m} \times 9.345 \times 10^{-5} = 812\ kPa$$

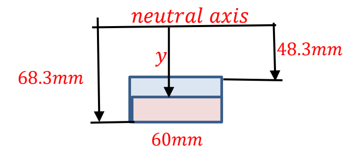
At the bottom of the web, $y = -0.0483m$ ⋙⋙ $A'\bar{y}' = 7.012 \times 10^{-5}$

$$\tau = \frac{V}{I_{zz}b}A'\bar{y}' = \frac{1.5 \times 10^3}{8.63 \times 10^{-6}\, m^4 \times 0.02\, m} \times 7.012 \times 10^{-5} = 609\, kPa$$

Shear Stress on the bottom flange

$A' = 0.06(0.0683 - y)$ ⋘⋙ $\bar{y}' = \frac{(0.0683 + y)}{2}$

$$A'\bar{y}' = 0.03(0.0683^2 - y^2)$$



At the bottom of the flange, $y = -0.0683\, m$.

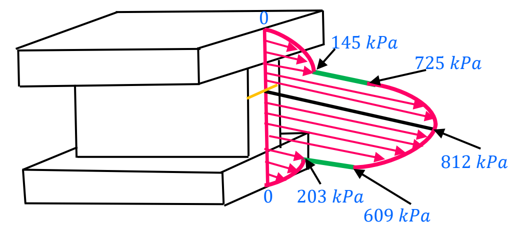
Therefore, $\tau = \frac{V}{I_{zz}b}A'\bar{y}' = 0$

At the joint with the web,

$y = -0.0483\, m$ ⋙⋙ $A'\bar{y}' = 6.996 \times 10^{-5}$

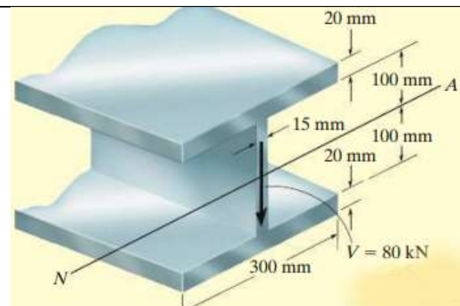
$$\tau = \frac{V}{I_{zz}b}A'\bar{y}' = \frac{1.5 \times 10^3}{8.63 \times 10^{-6}\, m^4 \times 0.06\, m} \times 6.996 \times 10^{-5}$$

$$= 203\, kPa$$



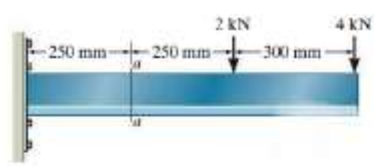
Exercise 1

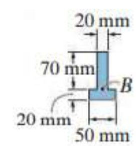
A steel wide flange beam has the dimensions as shown. It carries a transverse shear force $V = 80\, kN$. Plot the shear stress distribution across the cross section of the beam



Exercise 2

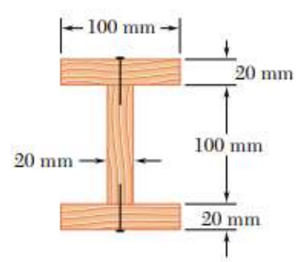
Determine the shear stress distribution across the cross section of the beam at location a-a.



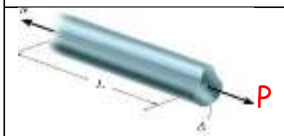
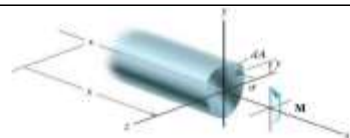
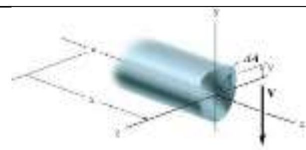


Exercise 3

A beam is made of three planks, $20mm x 100mm$ in cross section, nailed together. Knowing that the spacing between nails is $25\, mm$ and that the vertical shear in the beam is $V = 500\, N$. determine the shearing force in each nail.



Elastic Strain Energy for various types of loading

<table>
<tr><td rowspan="2">1. Axial</td><td>
</td><td>$U = \int_V \frac{\sigma_x{}^2}{2E}\, dV = \int_V \frac{P^2}{2EA^2}\, dV$
$U = \int_0^L \frac{P^2}{2AE}\, dx$</td></tr>
<tr><td>
</td><td>$U = \frac{P^2 L}{2AE}$</td></tr>
<tr><td>2. Torsion</td><td></td><td>$U = \int_V \frac{\tau^2}{2G}\, dV = \int_V \frac{1}{2G}\left(\frac{Tr}{J}\right)^2 dA\, dx$
$U = \int_V \frac{T^2}{2GJ^2}\left(\int r^2 dA\right) dx = \int_0^L \frac{T^2}{2GJ} dx$</td></tr>
<tr><td>3. Bending</td><td></td><td>$U = \int_V \frac{\sigma_x{}^2}{2E}\, dV = \int_V \frac{1}{2E}\left(\frac{My}{I}\right)^2 dA\, dx$
$U = \int_0^L \frac{M^2}{2EI^2}\left(\int y^2 dA\right) dx = \int_0^L \frac{M^2}{2EI} dx$</td></tr>
<tr><td rowspan="2">4. Transverse shear</td><td></td><td>$U = \int_V \frac{\tau^2}{2G}\, dV = \int_V \frac{1}{2G}\left(\frac{VQ}{It}\right)^2 dA\, dx$
$U = \int_0^L \frac{V^2}{2GI^2}\left(\int \frac{Q^2}{t^2} dA\right) dx$</td></tr>
<tr><td>Form factor
$f_s = \frac{A}{I^2}\int \frac{Q^2}{t^2} dA$</td><td>$U = \int_0^L \frac{f_s V^2}{2GA} dx$</td></tr>
</table>

References

1. Mechanics of Materials, 2/e, J. M. Gere & S. P.Timoshenko, CBS Publishers.
2. Mechanics of Materials, 6/e, F.P. Beer & E. R. Johnston, McGraw Hill
3. Engineering Mechanics of Solids, 2/e, E. P. Popov, Pearson Education Asia.
4. Mechanics of Materials, 9/e, R. C. Hibbeler, Prentice Hall.